

AWS Config + SNS: Real-Time S3 Public Access Monitoring

Technical Documentation &
Installation Guide

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Objective

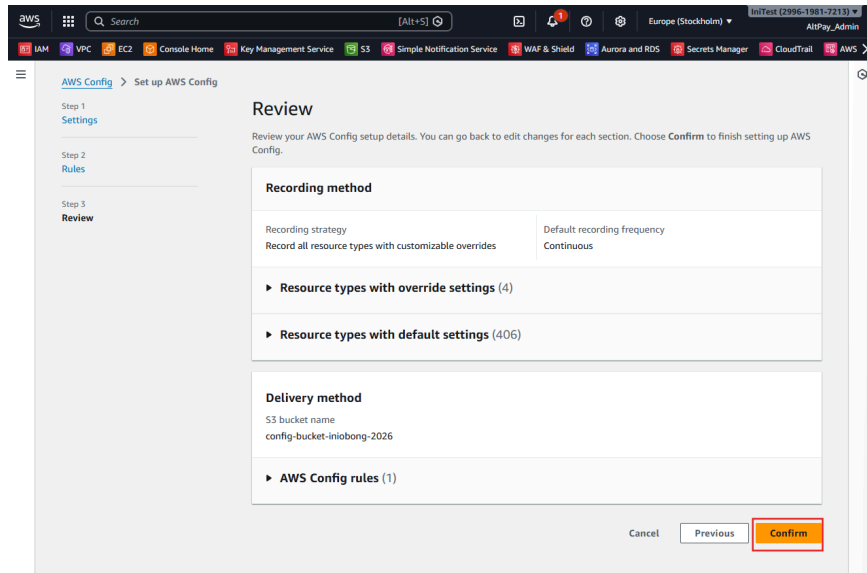
The objective of this task is to configure AWS Config and Amazon SNS to automatically monitor S3 bucket public access settings and send real time email alerts whenever an S3 bucket becomes publicly accessible, ensuring timely detection of non compliant configurations.

Step 1: Setup AWS Config

- On the AWS Management Console, search for AWS Config. Click on Get started
- You will be prompted to define the Recording method. For the Recording Strategy click on All resource types with customizable overrides.
- For the Default setting, stick with the default.
- Under the IAM role, select Create AWS Config
- Under Delivery channel, select Create a bucket, you can customise the S3 bucket name to your preference
- Leave the Amazon SNS topic setting as this will be created separately
- Click on Next
- Set Rule, search for **s3-bucket-public-read-prohibited**
- Click **Next**, then **Review and Confirm**

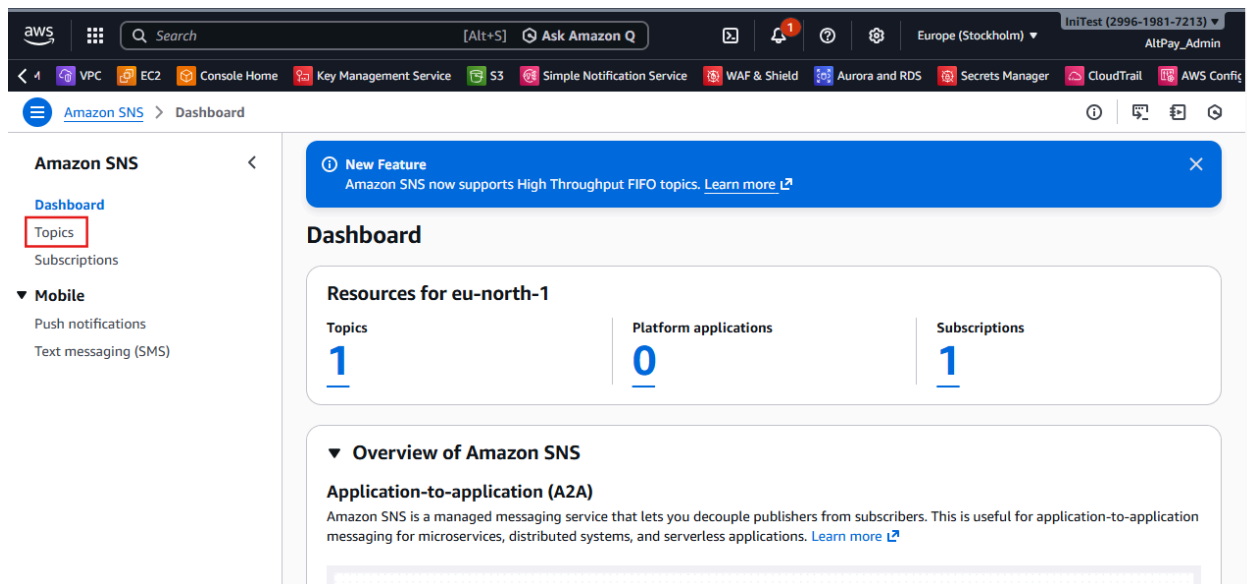
The screenshot shows the AWS Management Console interface for setting up AWS Config. The left sidebar indicates the current step is 'Step 1: Settings'. The main content area is titled 'Settings' and contains the following sections:

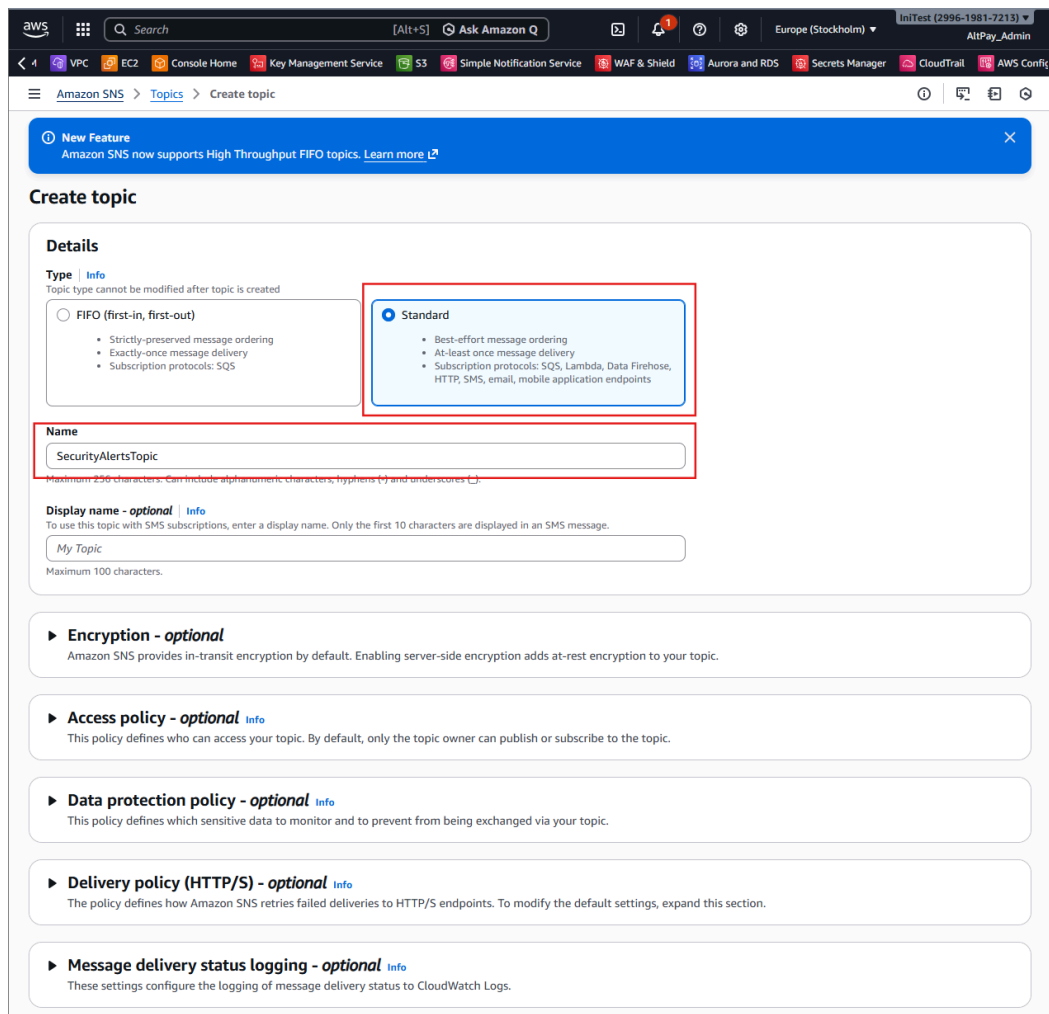
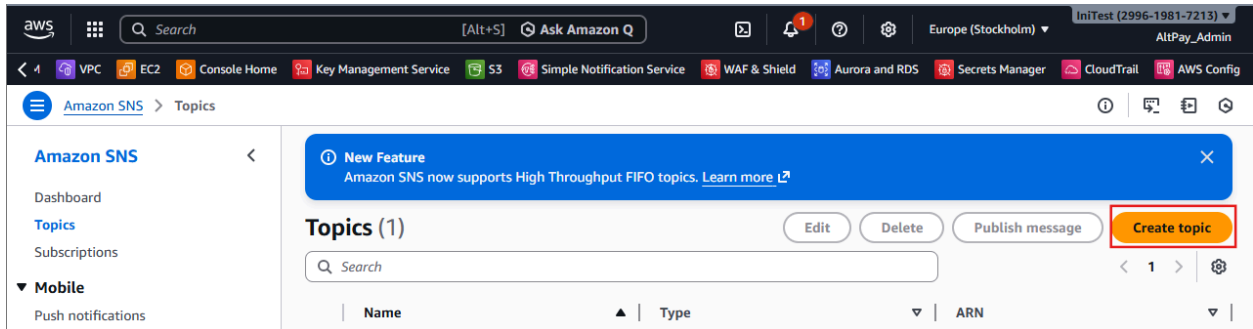
- Recording method**
 - Recording strategy**: Customise AWS Config to record configuration changes for all supported resource types, or for only the supported resource types that are relevant to you. Global resource types (RDS global clusters and IAM users, groups, roles, and customer managed policies) might be recorded in more than this Region. [Learn more](#) You are charged based on the number of configuration items recorded. [Pricing details](#)
 - ☒ **All resource types with customizable overrides**: AWS Config will record all current and future supported resource types in this Region. You can override the recording frequency for specific resource types or exclude specific resource types from recording.
 - ☐ **Specific resource types**: AWS Config will only record the resource types that you specify.
- Default settings**
 - Recording frequency**: Configure the default recording frequency for all current and future supported resource types. It impacts the cost to your bill. [Pricing details](#)
 - ☒ **Continuous recording**: Record configuration changes continuously whenever a change occurs.
 - ☐ **Daily recording**: Receive configuration data once every day only if a change has occurred.
- Override settings**
 - Resource types to override** [Info](#): Override the recording frequency for specific resource types, or exclude specific resource types from recording. If you change the recording frequency for a resource type or stop recording a resource type, the configuration items that were already recorded will remain unchanged.
 - A notification box states: "All globally recorded IAM resource types" are initially excluded from recording to help you reduce costs. Choose **Remove** to remove the override and include these resources in your recording.
 - A table with columns 'Resource type' and 'Override'. The first row shows 'All globally recorded IAM resource ...' with an 'Exclude from recording' override and a 'Remove' button.
 - An 'Add resource type' button.
 - A note: 'You can add up to 100 more frequency overrides and 596 more exclusion overrides.'



Step 2: Setup Amazon SNS

- Search for Amazon SNS on the AWS Management Console Search bar
- Click on Topic, then click on Create Topic
- Select Standard type and give your Topic a Name
- Click on Create topic





Step 3: Create Subscription

- Inside/within the newly created Topic, click on Create subscription
- Select a Protocol, I will be going for Email, under Endpoint include the recipient email you would like to receive the notifications
- Click on Create subscription
- Next, confirm Subscription. To confirm subscription, check your email/the email you entered when creating the subscription.

- Click on the Confirm subscription link (this enables you to review the notification alert).
- After confirmation refresh the subscription page on AWS, you should see the status change from Pending confirmation to Confirmed

The screenshot shows the Amazon SNS console interface. On the left, the navigation pane includes 'Amazon SNS', 'Dashboard', 'Topics', 'Subscriptions', and 'Mobile'. The main content area is titled 'SecurityAlertsTopic' and includes buttons for 'Edit', 'Delete', and 'Publish message'. Below this is a 'Details' section with a table containing the following information:

Name	ARN	Display name	Type
SecurityAlertsTopic	arn:aws:sns:eu-north-1:299619817213:SecurityAlertsTopic	-	Standard

Below the table, it indicates the 'Topic owner' as '299619817213'. At the bottom of the details section, there are tabs for 'Subscriptions', 'Access policy', 'Data protection policy', 'Delivery policy (HTTP/S)', and 'Delivery status log'. The 'Subscriptions' tab is active, showing a 'Subscriptions (0)' section with buttons for 'Edit', 'Delete', 'Request confirmation', 'Confirm subscription', and 'Create subscription'. A search bar and a table with columns 'ID', 'Endpoint', 'Status', and 'Protocol' are also present. A message states: 'No subscriptions found. You don't have any subscriptions to this topic.' with a 'Create subscription' button below it.

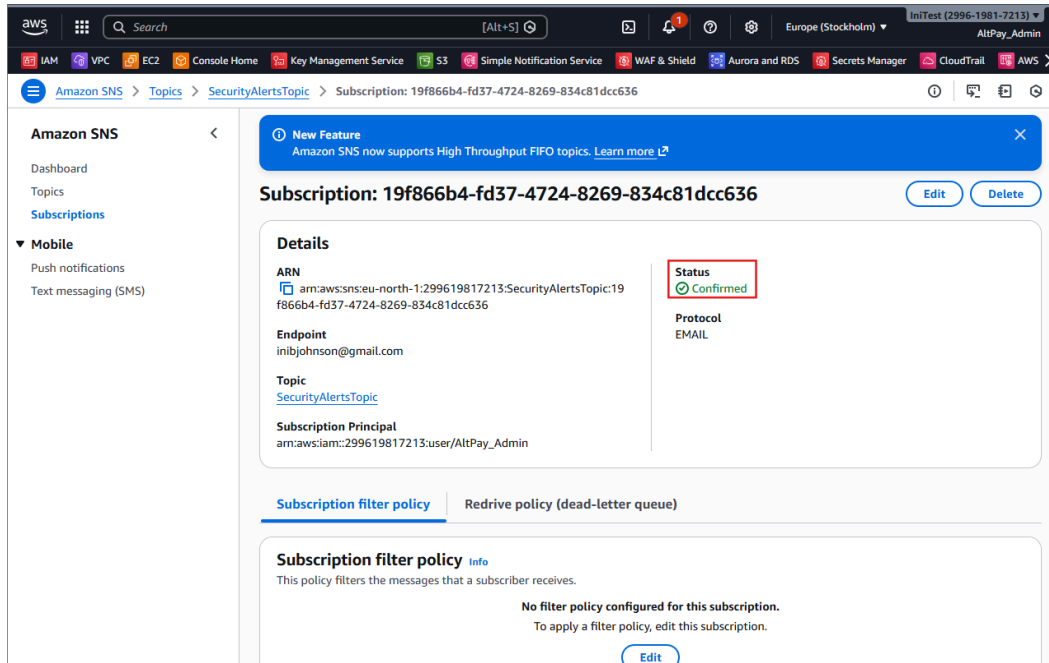
The screenshot shows the 'Create subscription' form in the Amazon SNS console. It includes a 'New Feature' banner at the top. The form has a 'Details' section with the following fields:

- Topic ARN:** A text field containing 'arn:aws:sns:eu-north-1:299619817213:SecurityAlertsTopic'.
- Protocol:** A dropdown menu with 'Email' selected.
- Endpoint:** A text field containing 'test@example.com'.

Below these fields, there is a note: 'After your subscription is created, you must confirm it. info'. Further down, there are two optional sections:

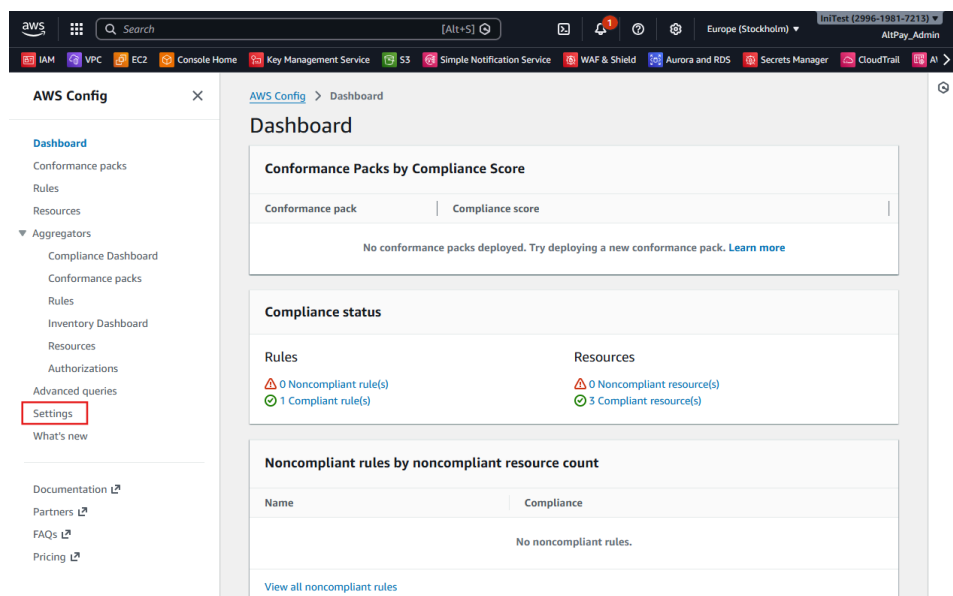
- Subscription filter policy - optional:** A section for filtering messages.
- Redrive policy (dead-letter queue) - optional:** A section for sending undeliverable messages to a dead-letter queue.

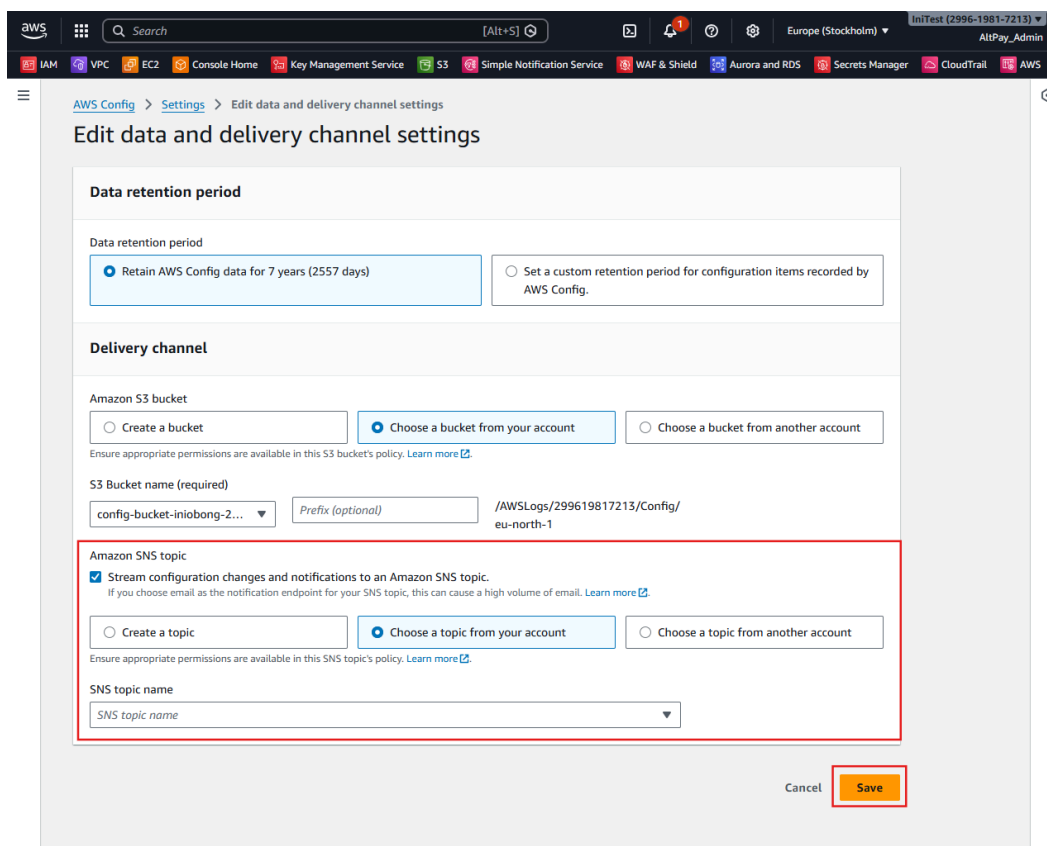
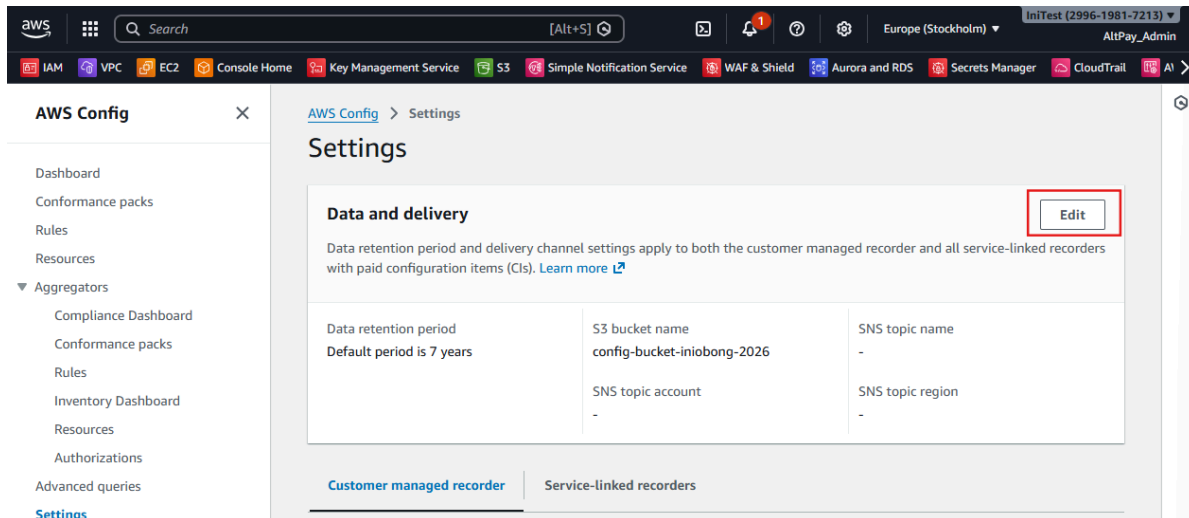
At the bottom right, there are 'Cancel' and 'Create subscription' buttons.



Step 4: Link the Create AWS Config to SNS

- On the AWS Config page, click on settings
- On the Data delivery click on Edit
- Under Amazon SNS topic click on the checkbox.
- Select the Choose a topic from your account.
- Select the newly created SNS topic name from the dropdown
- Click on Save





Step 5: Test the implementation

Firstly, break:

- **Action:** Create a Test S3 Bucket: Go to the S3 console, create a new bucket, and intentionally
- Change its permissions to make it publicly accessible.

- **Expected Result:** The new rule should be active. After a few minutes, the AWS Config Dashboard should show the S3-bucket-public-read-prohibited rule turning RED (Non-compliant)

NOTE: For there newly created bucket to trigger the notification, you need to give it a bucket policy. See below and replace YOUR-BUCKET-NAME with your actual bucket name.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadAccess",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

The screenshot shows the AWS Config Dashboard in the AWS Management Console. The left sidebar contains navigation links for various AWS services and Config-specific features. The main content area displays the 'Dashboard' with three sections:

- Conformance Packs by Compliance Score:** A table with columns 'Conformance pack' and 'Compliance score'. It shows 'No conformance packs deployed. Try deploying a new conformance pack. [Learn more](#)'
- Compliance status:** A summary section showing:

Rules	Resources
⚠️ 1 Noncompliant rule(s)	⚠️ 1 Noncompliant resource(s)
✅ 0 Compliant rule(s)	✅ 3 Compliant resource(s)
- Noncompliant rules by noncompliant resource count:** A table showing rules with non-compliant resources.

Name	Compliance
s3-bucket-public-read-prohi...	⚠️ 1 Noncompliant resource(s)

 Below the table is a link: [View all noncompliant rules](#)

AWS Config

Rules

A rule is a compliance check that helps you manage your ideal configuration settings. AWS Config evaluates whether your resource configurations comply with relevant rules and displays the compliance results.

Rules

Filter by compliance status: Noncompliant

Name	Remediation action	Type	Enabled evaluation...	Detective complia...
s3-bucket-public-r...	Not set	AWS managed	DETECTIVE	1 Noncompliant...

AWS Notification - Subscription Confirmation

AWS Notifications <no-reply@sns.amazonaws.com> to me

You have chosen to subscribe to the topic:
arn:aws:sns:eu-north-1:299619817213:SecurityAlertsTopic

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
[Confirm subscription](#)

Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to [sns-opt-out](#)

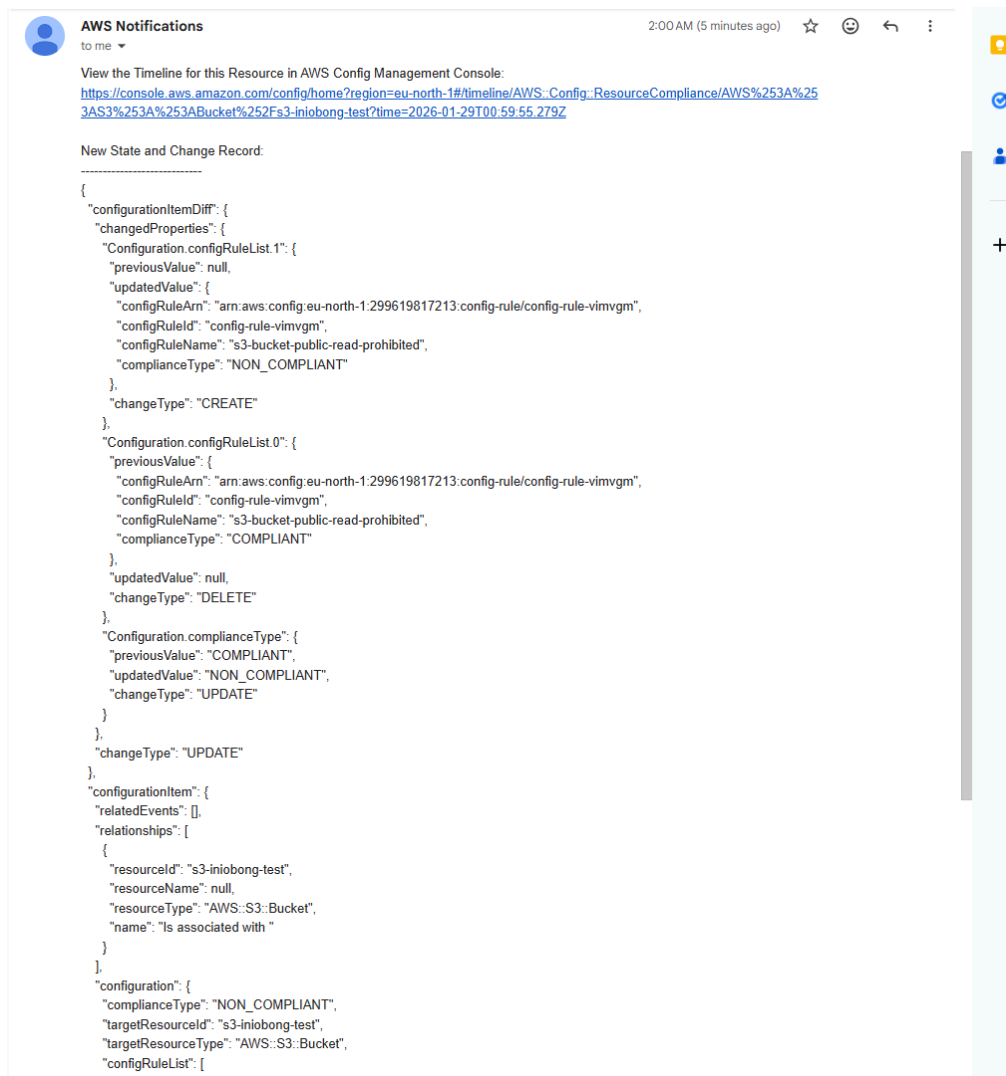
Primary

From	Subject	Time
Google	Security alert - A new sign-in on Windows inibjohnson@gmail.com We noticed a new sig...	2:04 AM
AWS Notifications 3	[AWS Config:eu-north-1] AWS::Config::ResourceCompliance AWS::S3::Bucket/s3-iniobon...	2:00 AM
AWS Notifications 2	[AWS Config:eu-north-1] AWS::S3::Bucket s3-iniobong-test is NON_COMPLIANT with s3-...	1:59 AM
AWS Notifications 3	[AWS Config:eu-north-1] AWS::S3::Bucket s3-iniobong-test Updated in Account 29961981...	1:59 AM
AWS Notifications 2	[AWS Config:eu-north-1] AWS::S3::Bucket s3-iniobong-test is COMPLIANT with s3-bucke...	1:57 AM
AWS Notifications	[AWS Config:eu-north-1] AWS::Config::ResourceCompliance AWS::S3::Bucket/s3-iniobon...	1:45 AM
AWS Notifications	[AWS Config:eu-north-1] AWS::S3::Bucket s3-iniobong-test Discovered in Account 29961...	1:44 AM
AWS Notifications	AWS Notification - Subscription Confirmation - You have chosen to subscribe to the topic...	1:09 AM

Secondly, Observe:

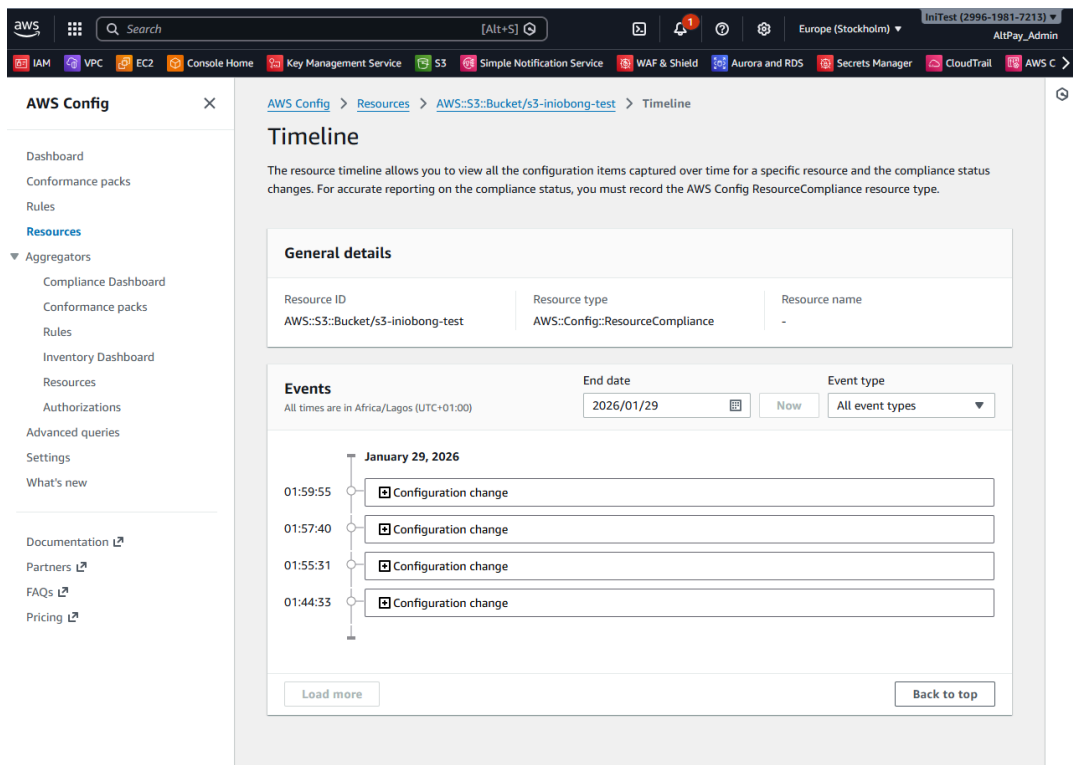
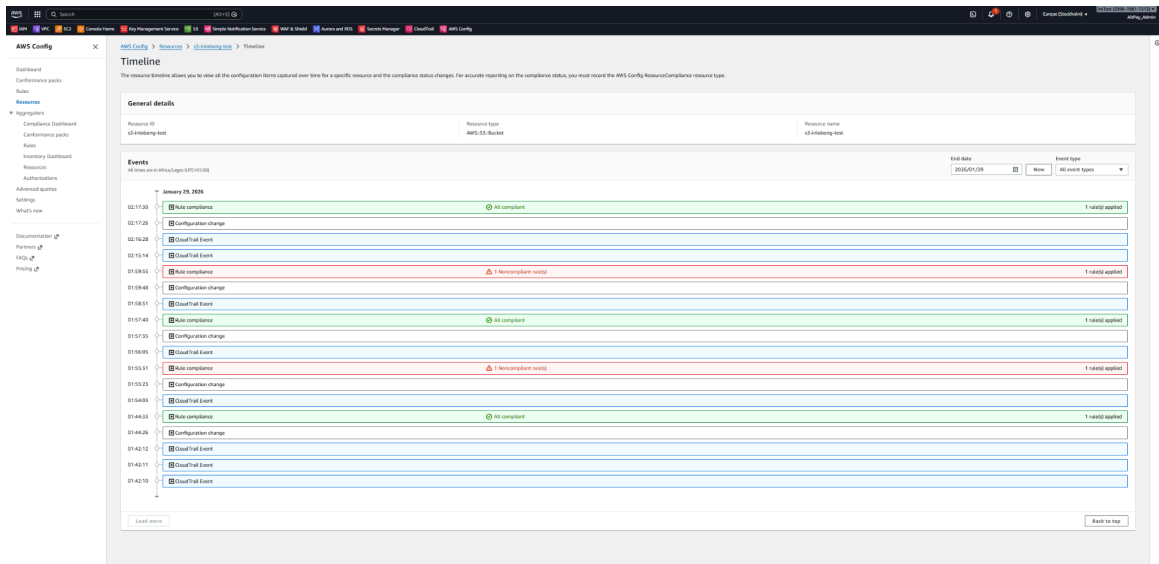
- **Action:** Check your Email: Wait for the alert to arrive in your personal inbox. The email will confirm the rule is non-compliant.

- **Observation:** You receive a real-time email notification from AWS SNS with the alert details.



Thirdly, Investigate:

- **Action:** View the Timeline: Go back to AWS Config, navigate to Resource inventory, find your test S3 bucket, and click on Resource timeline.
- **Observation:** You can see the exact time and Configuration change that made the bucket public.



Fourthly, fix:

- **Action:** Fix the Resource: Go back to your test S3 bucket in the S3 console and change its permissions back to private.
- **Observation:** After a few more minutes, the AWS Config Dashboard should return to GREEN (Compliant).

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 **AWS Notifications** to me 2:18 AM (6 minutes ago) ☆ 😊 ↶ ⋮

View the Timeline for this Resource in AWS Config Management Console:
<https://console.aws.amazon.com/config/home?region=eu-north-1#/timeline/AWS::Config::ResourceCompliance/AWS%253A%253AS3%253A%253ABucket%252Fs3-iniobong-test?time=2026-01-29T01:17:30.462Z>

New State and Change Record:

```
{
  "configurationItemDiff": {
    "changedProperties": {
      "Configuration.configRuleList.1": {
        "previousValue": null,
        "updatedAt": {
          "configRuleArn": "arn:aws:config:eu-north-1:299619817213:config-rule/config-rule-vimvgm",
          "configRuleId": "config-rule-vimvgm",
          "configRuleName": "s3-bucket-public-read-prohibited",
          "complianceType": "COMPLIANT"
        },
        "changeType": "CREATE"
      },
      "Configuration.configRuleList.0": {
        "previousValue": {
          "configRuleArn": "arn:aws:config:eu-north-1:299619817213:config-rule/config-rule-vimvgm",
          "configRuleId": "config-rule-vimvgm",
          "configRuleName": "s3-bucket-public-read-prohibited",
          "complianceType": "NON_COMPLIANT"
        },
        "updatedAt": null,
        "changeType": "DELETE"
      },
      "Configuration.complianceType": {
        "previousValue": "NON_COMPLIANT",
        "updatedAt": "COMPLIANT",
        "changeType": "UPDATE"
      }
    },
    "changeType": "UPDATE"
  },
  "configurationItem": {
    "relatedEvents": [],
    "relationships": [
      {
        "resourceId": "s3-iniobong-test",
        "resourceName": null,
        "resourceType": "AWS::S3::Bucket",
        "name": "Is associated with "
      }
    ]
  },
  "configuration": {
    "complianceType": "COMPLIANT",
    "targetResourceId": "s3-iniobong-test",
    "targetResourceType": "AWS::S3::Bucket",
    "configRuleList": [

```

NOTE: To fix it back, you need to either delete the JSON script or edit the policy to denie access.